10 Techniques

The Techniques chapter provides a high-level overview of the techniques referenced in the Knowledge Areas of the *BABOK® Guide*. Techniques are methods business analysts use to perform business analysis tasks.

The techniques described in the *BABOK® Guide* are intended to cover the most common and widespread techniques practiced within the business analysis community. Business analysts apply their experience and judgment in determining which techniques are appropriate to a given situation and how to apply each technique. This may include techniques that are not described in the *BABOK® Guide*. As the practice of business analysis evolves, techniques will be added, changed, or removed from future iterations of the *BABOK® Guide*.

In a number of cases, a set of conceptually similar approaches have been grouped into a single technique. Any approach within a technique may be used individually or in combination to accomplish the technique's purpose.

10.1 Acceptance and Evaluation Criteria

10.1.1 Purpose

Acceptance criteria are used to define the requirements, outcomes, or conditions that must be met in order for a solution to be considered acceptable to key stakeholders. Evaluation criteria are the measures used to assess a set of requirements in order to choose between multiple solutions.

10.1.2 Description

Acceptance and evaluation criteria define measures of value attributes to be used for assessing and comparing solutions and alternative designs. Measurable and testable criteria allow for the objective and consistent assessment of solutions and designs. The Acceptance and Evaluation Criteria technique can apply at all levels of a project, from high-level to a more detailed level.

Acceptance criteria describe the minimum set of requirements that must be met in order for a particular solution to be worth implementing. They may be used to determine if a solution or solution component can meet a requirement. Acceptance criteria are typically used when only one possible solution is being evaluated, and are generally expressed as a pass or fail.

Evaluation criteria define a set of measurements which allow for ranking of solutions and alternative designs according to their value for stakeholders. Each evaluation criterion represents a continuous or discrete scale for measuring a specific solution attribute such as cost, performance, usability, and how well the functionality represents the stakeholders' needs. Attributes that cannot be measured directly are evaluated using expert judgment or various scoring techniques.

Both evaluation and acceptance criteria may be defined with the same value attributes. When evaluating various solutions, the solutions with lower costs and better performance may be rated higher. When accepting a solution, the criteria are written using minimum performance requirements and maximum cost limits in contractual agreements and user acceptance tests.

10.1.3 Elements

.1 Value Attributes

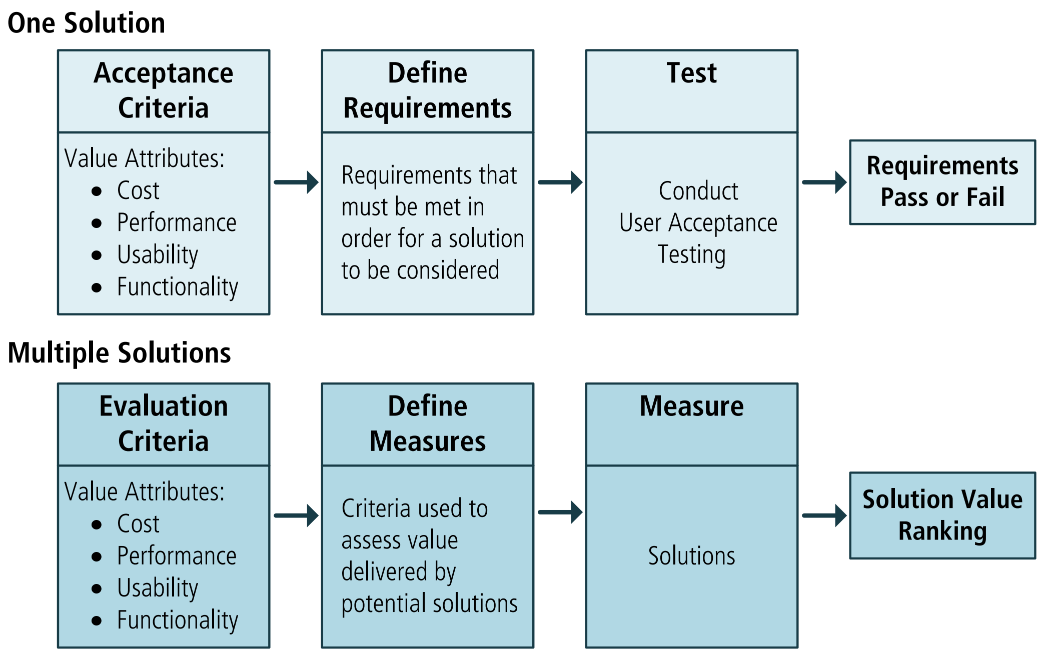
Value attributes are the characteristics of a solution that determine or substantially influence its value for stakeholders. They represent a meaningful and agreed-upon decomposition of the value proposition into its constituent parts, which can be described as qualities that the solution should either possess or avoid.

Examples of value attributes include:

- ability to provide specific information,
- ability to perform or support specific operations,
- performance and responsiveness characteristics,
- applicability of the solution in specific situations and contexts,
- availability of specific features and capabilities, and
- usability, security, scalability, and reliability.

Basing acceptance and evaluation criteria on value attributes ensures that they are valid and relevant to stakeholder needs and should be considered when accepting and evaluating the solution. Business analysts ensure that the definition of all value attributes are agreed upon by all stakeholders. Business analysts may design tools and instructions for performing the assessment as well as for recording and processing its results.

Figure 10.1.1: Acceptance and Evaluation Criteria



.2 Assessment

In order to assess a solution against acceptance or evaluation criteria, it must be constructed in a measurable format.

Testability

Acceptance criteria are expressed in a testable form. This may require breaking requirements down into an atomic form so that test cases can be written to verify the solution against the criteria. Acceptance criteria are presented in the form of statements which can be verified as true or false. This is often achieved through user acceptance testing (UAT).

Measures

Evaluation criteria provide a way to determine if features provide the value necessary to satisfy stakeholder needs. The criteria are presented as parameters that can be measured against a continuous or discrete scale. The definition of each criterion allows the solution to be measured through various methods such as benchmarking or expert judgment. Defining evaluation criteria may involve designing tools and instructions for performing the assessment, as well as for recording and processing its results.

10.1.4 Usage Considerations

.1 Strengths

- Agile methodologies may require that all requirements be expressed in the form of testable acceptance criteria.
- Acceptance criteria are necessary when the requirements express contractual obligations.
- Acceptance criteria provide the ability to assess requirements based on agreed-upon criteria.
- Evaluation criteria provide the ability to assess diverse needs based on agreed-upon criteria, such as features, common indicators, local or global benchmarks, and agreed ratios.
- Evaluation criteria assist in the delivery of expected return on investment (ROI) or otherwise specified potential value.
- Evaluation criteria helps in defining priorities.

.2 Limitations

- Acceptance criteria may express contractual obligations and as such may be difficult to change for legal or political reasons.
- Achieving agreement on evaluation criteria for different needs among diverse stakeholders can be challenging.

10.2 Backlog Management

10.2.1 Purpose

The backlog is used to record, track, and prioritize remaining work items.

10.2.2 Description

A backlog occurs when the volume of work items to be completed exceeds the capacity to complete them.

Backlog management refers to the planned approach to determine:

- what work items should be formally included in the backlog,
- how to describe the work items,
- how the work items should be tracked,
- how the work items should be periodically reviewed and prioritized in relation to all other items in the backlog,
- how the work items are eventually selected to be worked on, and
- how the work items are eventually removed from the backlog.

In a managed backlog, the items at the top have the highest business value and the highest priority. These are normally the next items to be selected to be worked on.

Periodic review of the entire backlog should occur because changes in stakeholder needs and priorities may necessitate changes to the priority of some of the backlog items. In many environments, the backlog is reviewed at planned intervals.

The changes to the number of items in the backlog are regularly monitored. The root causes for these changes are investigated: a growing backlog could indicate an increase in demand or a drop in productivity; a declining backlog could indicate a drop in demand or improvements in the production process.

There may be more than one backlog. For example, one backlog may be used to manage a global set of items, while a second backlog may be used to manage the items that are due to be worked on within the very near future.

10.2.3 Elements

.1 Items in the Backlog

Backlog items may be any kind of item which may have work associated with it. A backlog may contain, but is not limited to, any combination of the following items:

- use cases,
- user stories,
- functional requirements,
- non-functional requirements,
- designs,
- customer orders,
- risk items,
- change requests,
- defects,
- planned rework,
- maintenance,
- conducting a presentation, or
- completing a document.

An item is added to the backlog if it has value to a stakeholder. There may be one person with the authority to add new items to the backlog, or there could be a committee which adds new items based on a consensus. In some cases, the responsibility for adding new items may be delegated to the business analyst. There may also be policies and rules which dictate what is to be added and when, as may be the case with major product defects.

.2 Prioritization

Items in the backlog are prioritized relative to each other. Over time, these priorities will change as stakeholders' priorities change, or as dependencies between backlog items emerge. Rules on how to manage the backlog may also impact priority.

A multi-phased prioritization approach can also be used. When items are first added to the backlog, the prioritization may be very broad, using categories such as high, medium, or low. The high priority items tend to be reviewed more frequently since they are likely candidates for upcoming work. To differentiate between the high priority items, a more granular approach is used to specify the relative priority to other high priority items, such as a numerical ranking based on some measure of value.

.3 Estimation

The level of detail used to describe each backlog item may vary considerably. Items near the top of the backlog are usually described in more detail, with a correspondingly accurate estimate about their relative size and complexity that would help to determine the cost and effort to complete them. When an item is first added, there may be very little detail included, especially if the item is not likely to be worked on in the near term.

A minimal amount of work is done on each item while it is on the backlog; just enough to be able to understand the work involved to complete it. As the work progresses on other items in the backlog, an individual item's relative priority may rise, leading to a need to review it and possibly further elaborate or decompose it to better understand and estimate its size and complexity.

Feedback from the production process about the cost and effort to complete earlier items can be used to refine the estimates of items still in the backlog.

.4 Managing Changes to the Backlog

Items make their way to the top of the backlog based on their relative priority to other items in the backlog. When new or changed requirements are identified, they are added to the backlog and ordered relative to the other items already there.

Whenever work capacity becomes available the backlog is reviewed and items are selected based on the available capacity, dependencies between items, current understanding of the size, and complexity.

Items are removed from the backlog when they are completed, or if a decision has been made to not do any more work on them. However, removed items can be re-added to the backlog for a variety of reasons, including:

- stakeholder needs could change significantly,
- it could be more time-consuming than estimated,
- other priority items could take longer to complete than estimated, or
- the resulting work product might have defects.

10.2.4 Usage Considerations

.1 Strengths

- An effective approach to responding to changing stakeholder needs and priorities because the next work items selected from the backlog are always aligned with current stakeholder priorities.

- Only items near the top of the backlog are elaborated and estimated in detail; items near the bottom of the backlog reflect lower priorities and receive less attention and effort.
- Can be an effective communication vehicle because stakeholders can understand what items are about to be worked on, what items are scheduled farther out, and which ones may not be worked on for some time.

.2 Limitations

- Large backlogs may become cumbersome and difficult to manage.
- It takes experience to be able to break down the work to be done into enough detail for accurate estimation.
- A lack of detail in the items in the backlog can result in lost information over time.

10.3 Balanced Scorecard

10.3.1 Purpose

The balanced scorecard is used to manage performance in any business model, organizational structure, or business process.

10.3.2 Description

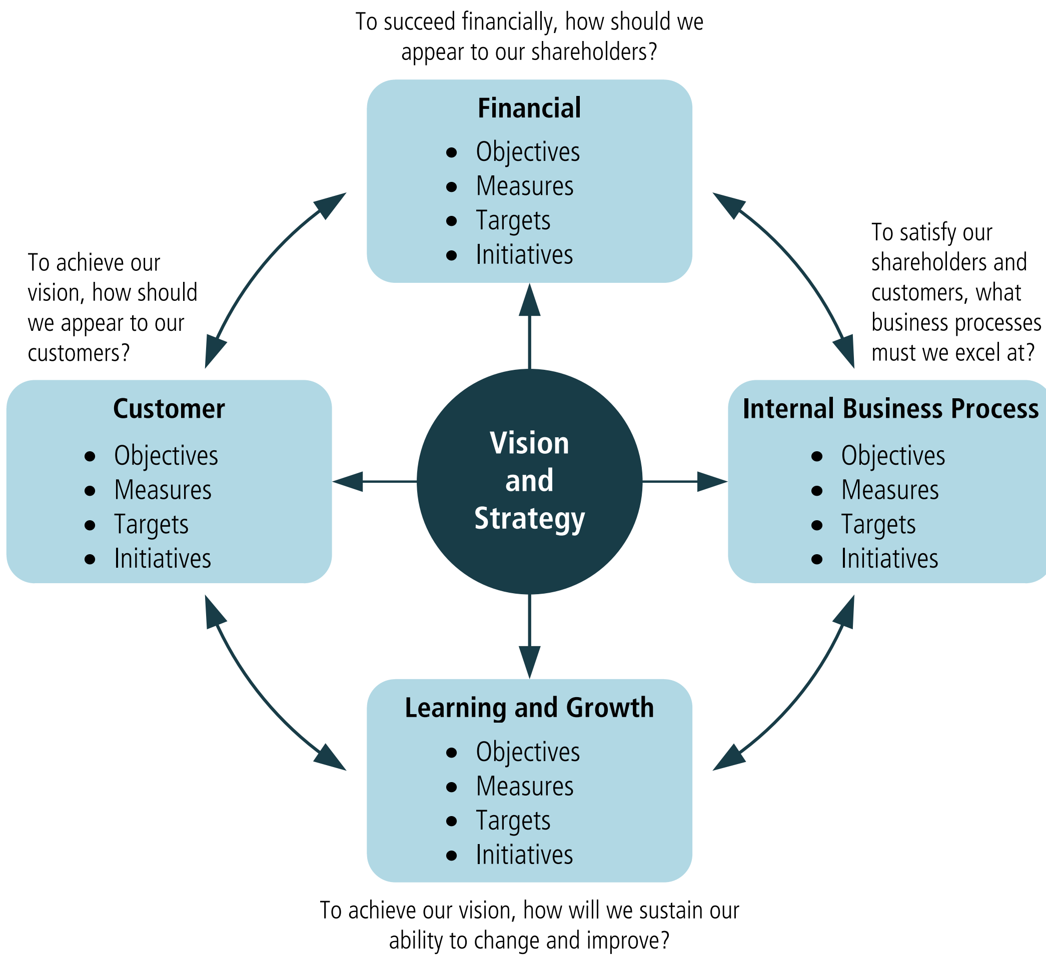
The balanced scorecard is a strategic planning and management tool used to measure organizational performance beyond the traditional financial measures. It is outcome focused and provides a balanced view of an enterprise by implementing the strategic plan as an active framework of objectives and performance measures. The underlying premise of the balanced scorecard is that the drivers of value creation are understood, measured, and optimized in order to create sustainable performance.

The balanced scorecard is composed of four dimensions:

- Learning and Growth,
- Business Process,
- Customer, and
- Financial.

The balanced scorecard includes tangible objectives, specific measures, and targeted outcomes derived from an organization's vision and strategy. Balanced business scorecards can be used at multiple levels within an organization. This includes at an enterprise-wide level (macro level), departmental or function level, and even at the level of a project or initiative.

Figure 10.3.1: Balanced Scorecard



10.3.3 Elements

.1 Learning and Growth Dimension

The Learning and Growth dimension includes measures regarding employee training and learning, product and service innovation, and corporate culture. Metrics guide the use of training funds, mentoring, knowledge sharing, and technology improvements.

.2 Business Process Dimension

The Business Process dimension includes metrics that indicate how well the enterprise is operating and if their products meet customer needs.

.3 Customer Dimension

The Customer dimension includes metrics on customer focus, satisfaction and delivery of value. These metrics capture how well customer needs are met, how satisfied they are with products and services, whether the delivery of those products and services meet their quality expectations, and their overall experience with the enterprise.

.4 Financial Dimension

The Financial dimension identifies what is financially necessary to realize the strategy. Examples of financial measures indicate profitability, revenue growth, and added economic value.

.5 Measures or Indicators

There are two basic types of measures or indicators: lagging indicators that provide results of actions already taken and leading indicators that provide information about future performance.

Objectives tend to have lagging indicators, but using related leading indicators can provide more real-time performance information.

10.3.4 Usage Considerations

In order for measures to be meaningful they should be quantitative, linked to strategy, and easily understood by all stakeholders. When defining measures, business analysts consider other relevant measures that are in place and ensure that any new or changed measures do not adversely impact any existing ones. At any time, any dimension of the balanced scorecard may be active, changing, and evolving. Each dimension affects and is affected by the others. The balanced scorecard allows the organization to establish monitoring and measuring of progress against objectives and to adapt strategy as needed.

Because scorecards are used to assess the performance of the enterprise or a business unit within the enterprise, changes to the measures can have wide reaching implications and must be clearly communicated and carefully managed.

.1 Strengths

- Facilitates holistic and balanced planning and thinking.
- Short-, medium-, and long-term goals can be harmonized into programs with incremental success measures.
- Strategic, tactical, and operational teams are more easily aligned in their work.
- Encourages forward thinking and competitiveness.

.2 Limitations

- A lack of a clear strategy makes aligning the dimensions difficult.
- Can be seen as the single tool for strategic planning rather than just one tool to be used in a suite of strategic planning tools.
- Can be misinterpreted as a replacement for strategic planning, execution, and measurement.

10.4 Benchmarking and Market Analysis

10.4.1 Purpose

Benchmarking and market analysis are conducted to improve organizational operations, increase customer satisfaction, and increase value to stakeholders.

10.4.2 Description

Benchmark studies are conducted to compare organizational practices against the best-in-class practices. Best practices may be found in competitor enterprises, in government, or from industry associations. The objective of benchmarking is to evaluate enterprise performance and ensure that the enterprise is operating efficiently. Benchmarking may also be performed against standards for compliance purposes. The results from the benchmark study may initiate change within an organization.

Market analysis involves researching customers in order to determine the products and services that they need or want, the factors that influence their decisions to purchase, and the competitors that exist in the market. The objective of market analysis is to acquire this information in order to support the various decision-making processes within an organization. Market analysis can also help determine when to exit a market. It may be used to determine if partnering, merging, or divesting are viable alternatives for an enterprise.

10.4.3 Elements

.1 Benchmarking

Benchmarking includes:

- identifying the areas to be studied,
- identifying enterprises that are leaders in the sector (including competitors),
- conducting a survey of selected enterprises to understand their practices,
- using a Request for Information (RFI) to gather information about capabilities,
- arranging visits to best-in-class organizations,
- determining gaps between current and best practices, and
- developing a project proposal to implement best practices.

.2 Market Analysis

Market Analysis requires that business analysts:

- identify customers and understand their preferences,
- identify opportunities that may increase value to stakeholders,
- identify competitors and investigate their operations,

- look for trends in the market, anticipate growth rate, and estimate potential profitability,
- define appropriate business strategies,
- gather market data,
- use existing resources such as company records, research studies, and books and apply that information to the questions at hand, and
- review data to determine trends and draw conclusions.

10.4.4 Usage Considerations

.1 Strengths

- Benchmarking provides organizations with information about new and different methods, ideas, and tools to improve organizational performance.
- An organization may use benchmarking to identify best practices by its competitors in order to meet or exceed its competition.
- Benchmarking identifies why similar companies are successful and what processes they used to become successful.
- Market analysis can target specific groups and can be tailored to answer specific questions.
- Market analysis may expose weaknesses within a certain company or industry.
- Market analysis may identify differences in product offerings and services that are available from a competitor.

.2 Limitations

- Benchmarking is time-consuming; organizations may not have the expertise to conduct the analysis and interpret useful information.
- Benchmarking cannot produce innovative solutions or solutions that will produce a sustainable competitive advantage because it involves assessing solutions that have been shown to work elsewhere with the goal of reproducing them.
- Market analysis can be time-consuming and expensive, and the results may not be immediately available.
- Without market segmentation, market analysis may not produce the expected results or may provide incorrect data about a competitor's products or services.

10.5 Brainstorming

10.5.1 Purpose

Brainstorming is an excellent way to foster creative thinking about a problem. The aim of brainstorming is to produce numerous new ideas, and to derive from them themes for further analysis.

10.5.2 Description

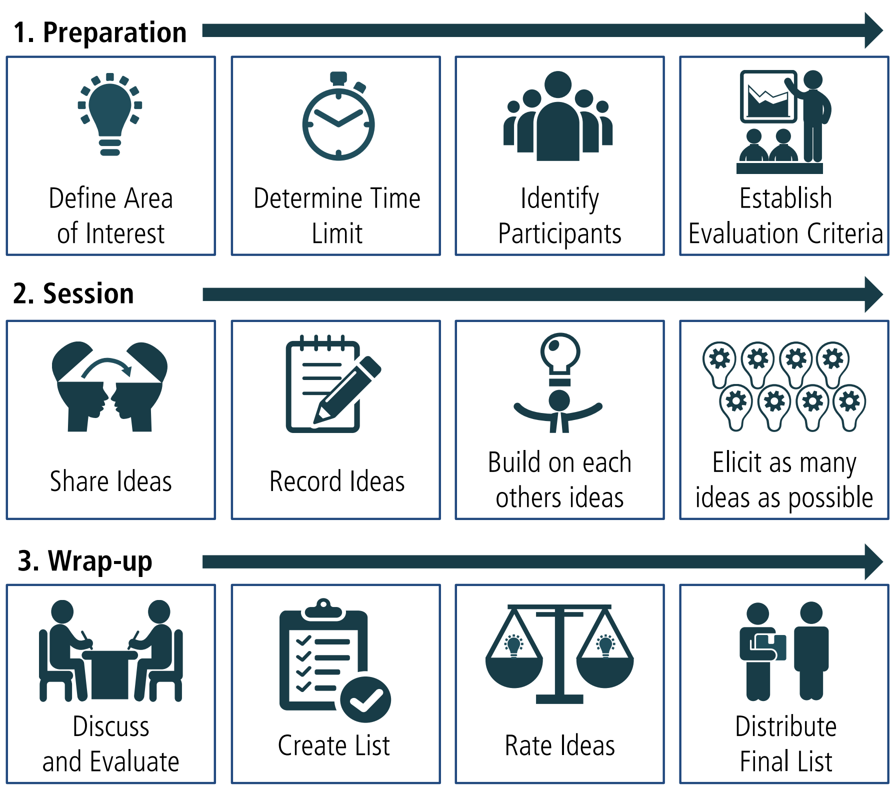
Brainstorming is a technique intended to produce a broad or diverse set of options.

It helps answer specific questions such as (but not limited to):

- What options are available to resolve the issue at hand?
- What factors are constraining the group from moving ahead with an approach or option?
- What could be causing a delay in activity 'A'?
- What can the group do to solve problem 'B'?

Brainstorming works by focusing on a topic or problem and then coming up with many possible solutions to it. This technique is best applied in a group as it draws on the experience and creativity of all members of the group. In the absence of a group, one could brainstorm on one's own to spark new ideas. To heighten creativity, participants are encouraged to use new ways of looking at things and freely associate in any direction. When facilitated properly, brainstorming can be fun, engaging, and productive.

Figure 10.5.1: Brainstorming



10.5.3 Elements

.1 Preparation

- Develop a clear and concise definition of the area of interest.
- Determine a time limit for the group to generate ideas; the larger the group, the more time required.
- Identify the facilitator and participants in the session (aim for six to eight participants who represent a range of backgrounds and experience with the topic).
- Set expectations with participants and get their buy-in to the process.
- Establish the criteria for evaluating and rating the ideas.

.2 Session

- Share new ideas without any discussion, criticism, or evaluation.
- Visibly record all ideas.
- Encourage participants to be creative, share exaggerated ideas, and build on the ideas of others.
- Don't limit the number of ideas as the goal is to elicit as many as possible within the time period.

.3 Wrap-up

- Once the time limit is reached, discuss and evaluate the ideas using the predetermined evaluation criteria.
- Create a condensed list of ideas, combine ideas where appropriate, and eliminate duplicates.
- Rate the ideas, and then distribute the final list of ideas to the appropriate parties.

10.5.4 Usage Considerations

.1 Strengths

- Ability to elicit many ideas in a short time period.
- Non-judgmental environment enables creative thinking.
- Can be useful during a workshop to reduce tension between participants.

.2 Limitations

- Participation is dependent on individual creativity and willingness to participate.
- Organizational and interpersonal politics may limit overall participation.
- Group participants must agree to avoid debating the ideas raised during brainstorming.

10.6 Business Capability Analysis

10.6.1 Purpose

Business capability analysis provides a framework for scoping and planning by generating a shared understanding of outcomes, identifying alignment with strategy, and providing a scope and prioritization filter.

10.6.2 Description

Business capability analysis describes what an enterprise, or part of an enterprise, is able to do. Business capabilities describe the ability of an enterprise to act on or transform something that helps achieve a business goal or objective. Capabilities may be assessed for performance and associated risks to identify specific performance gaps and prioritize investments. Many product development efforts are an attempt to improve the performance of an existing business capability or to deliver a new one. As long as an enterprise continues to perform similar functions, the capabilities required by the enterprise should remain constant—even if the method of execution for those capabilities undergoes significant change.

10.6.3 Elements

.1 Capabilities

Capabilities are the abilities of an enterprise to perform or transform something that helps achieve a business goal or objective. Capabilities describe the purpose or outcome of the performance or transformation, not how the performance or transformation is performed. Each capability is found only once on a capability map, even if it is possessed by multiple business units.

.2 Using Capabilities

Capabilities impact value through increasing or protecting revenue, reducing or preventing cost, improving service, achieving compliance, or positioning the company for the future. Not all capabilities have the same level of value. There are various tools that can be used to make value explicit in a capability assessment.

.3 Performance Expectations

Capabilities can be assessed to identify explicit performance expectations. When a capability is targeted for improvement, a specific performance gap can be identified. The performance gap is the difference between the current performance and the desired performance, given the business strategy.

.4 Risk Model

Capabilities alone do not have risks—the risks are in the performance of the capability, or in the lack of performance.

These risks fall into the usual business categories:

- business risk,
- technology risk,
- organizational risk, and
- market risk.

.5 Strategic Planning

Business capabilities for the current state and future state of an enterprise can be used to determine where that enterprise needs to go in order to accomplish its strategy. A business capability assessment can produce a set of recommendations or proposals for solutions. This information forms the basis of a product roadmap and serves as a guide for release planning. At the strategic level, capabilities should support an enterprise in establishing and maintaining a sustainable competitive advantage and a distinct value proposition.

.6 Capability Maps

Capability maps provide a graphical view of elements involved in business capability analysis. The following examples demonstrate one element of a capability map that would be part of a larger capabilities grid.

There is no set standard for the notation of capabilities maps. The following images show two different methods for creating a capability map. The first two images are the first example and the third image is the second example.

Figure 10.6.1: Sample Capability Map Example 1 Cell

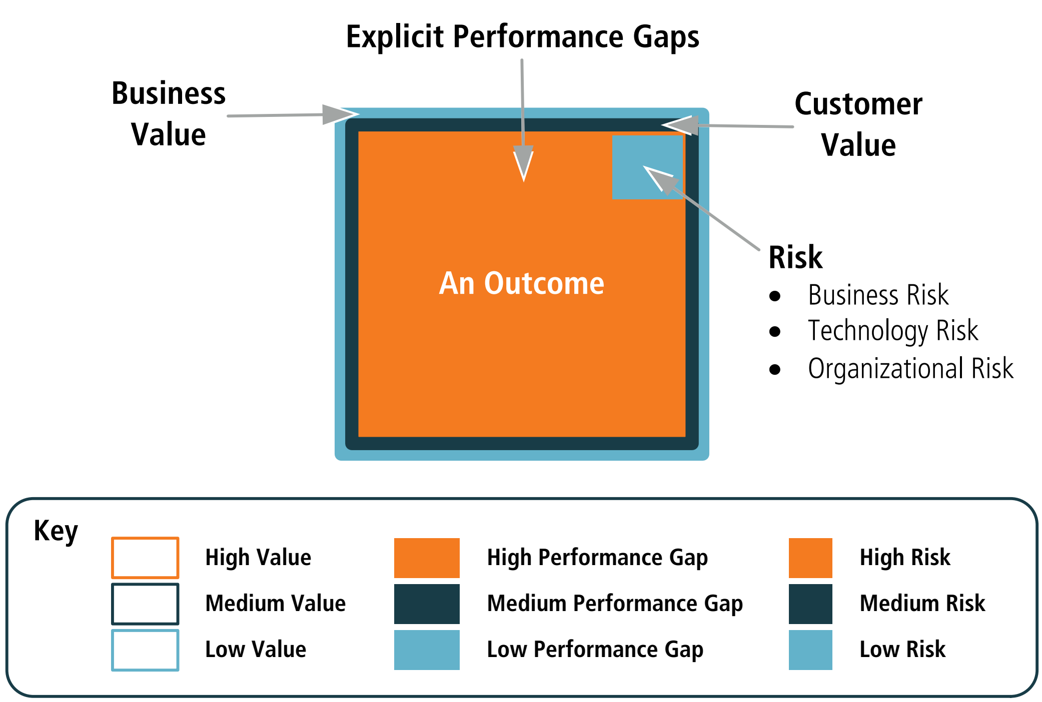
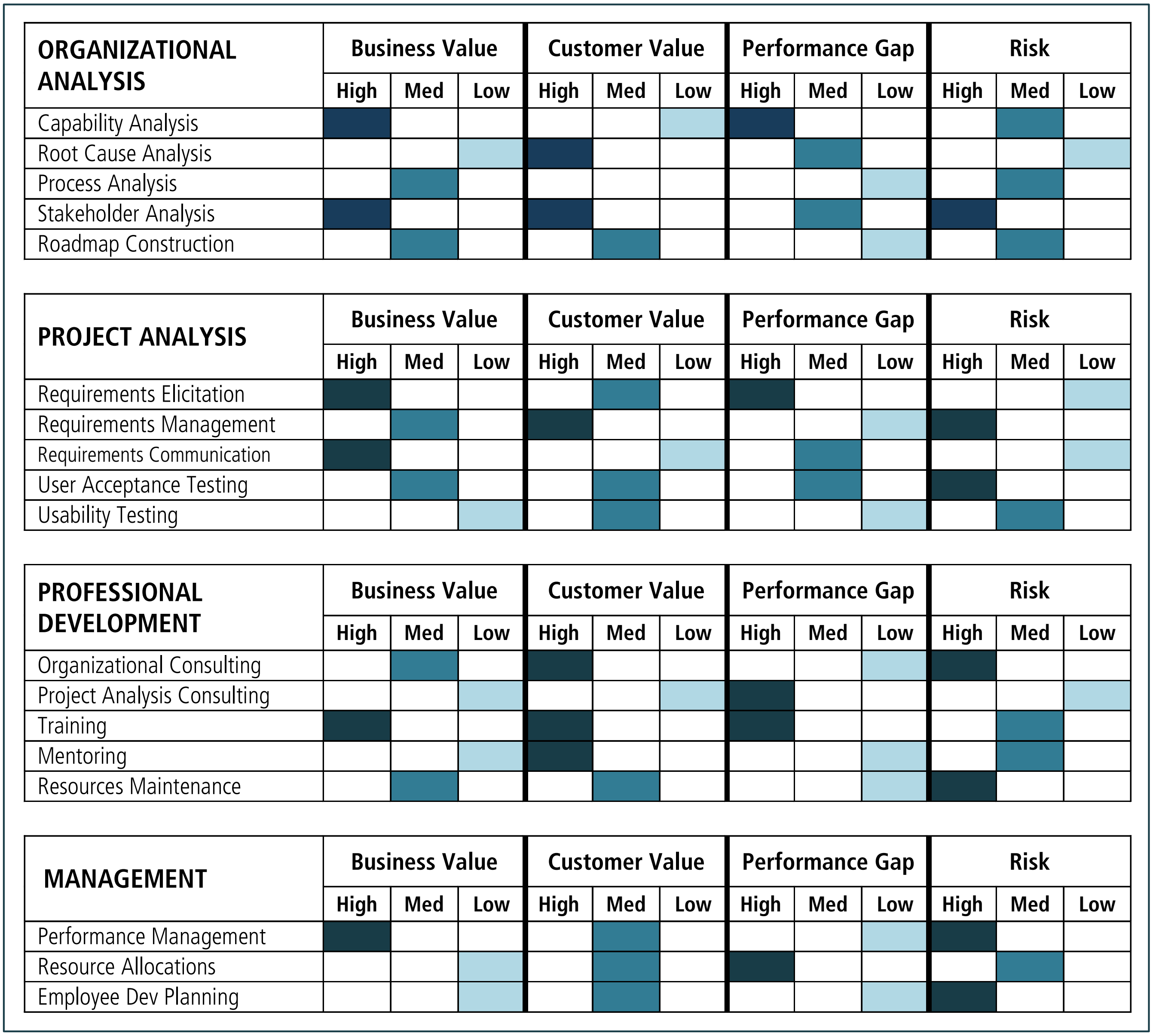


Figure 10.6.2: Sample Capability Map Example 1

Figure 10.6.3: Sample Capability Map Example 2



10.6.4 Usage Considerations

.1 Strengths

- Provides a shared articulation of outcomes, strategy, and performance, which help create very focused and aligned initiatives.
- Helps align business initiatives across multiple aspects of the organization.
- Useful when assessing the ability of an organization to offer new products and services.

.2 Limitations

- Requires an organization to agree to collaborate on this model.

- When created unilaterally or in a vacuum it fails to deliver on the goals of alignment and shared understanding.
- Requires a broad, cross–functional collaboration in defining the capability model and the value framework.

10.7 Business Cases

10.7.1 Purpose

A business case provides a justification for a course of action based on the benefits to be realized by using the proposed solution, as compared to the cost, effort, and other considerations to acquire and live with that solution.

10.7.2 Description

A business case captures the rationale for undertaking a change. A business case is frequently presented in a formal document, but may also be presented through informal methods. The amount of time and resources spent on the business case should be proportional to the size and importance of its potential value. The business case provides sufficient detail to inform and request approval without providing specific intricacies about the method and/or approach to the implementation. It may also be the catalyst for one or many initiatives in order to implement the change.

A business case is used to:

- define the need,
- determine the desired outcomes,
- assess constraints, assumptions, and risks, and
- recommend a solution.

10.7.3 Elements

.1 Need Assessment

The need is the driver for the business case. It is the relevant business goal or objective that must be met. Objectives are linked to a strategy or the strategies of the enterprise. The need assessment identifies the problem or the potential opportunity. Throughout the development of the business case, different alternatives to solve the problem or take advantage of the opportunity will be assessed.

.2 Desired Outcomes

The desired outcomes describe the state which should result if the need is fulfilled. They should include measurable outcomes that can be utilized to determine the success of the business case or the solution. Desired outcomes should be revisited at defined milestones and at the completion of the initiative (or initiatives) to fulfill the business case. They should also be independent of the recommended solution. As solution options are assessed, their ability to achieve the desired outcomes will help determine the recommended solution.

.3 Assess Alternatives

The business case identifies and assesses various alternative solutions. Alternatives may include (but are not limited to) different technologies, processes, or business models. Alternatives may also include different ways of acquiring these and different timing options. They will be affected by constraints such as budget, timing, and regulatory. The 'do-nothing' alternative should be assessed and considered for the recommended solution.

Each alternative should be assessed in terms of:

- **Scope**: defines the alternative being proposed. Scope can be defined using organizational boundaries, system boundaries, business processes, product lines or geographic regions. Scope statements clearly define what will be included and what will be excluded. The scope of various alternatives may be similar or have overlap but may also differ based on the alternative.
- **Feasibility**: The organizational and technical feasibility should be assessed for each alternative. It includes organizational knowledge, skills, and capacity, as well as technical maturity and experience in the proposed technologies.
- **Assumptions, Risks, and Constraints**: Assumptions are agreed-to facts that may have influence on the initiative. Constraints are limitations that may restrict the possible alternatives. Risks are potential problems that may have a negative impact on the solution. Agreeing to and documenting these factors facilitates realistic expectations and a shared understanding amongst stakeholders.
- **Financial Analysis and Value Assessment**: The financial analysis and value assessment includes an estimate of the costs to implement and operate the alternative, as well as a quantified financial benefit from implementing the alternative. Benefits of a non-financial nature (such as improved staff morale, increased flexibility to respond to change, improved customer satisfaction, or reduced exposure to risk) are also important and add significant value to the organization. Value estimates are related back to strategic goals and objectives.

.4 Recommended Solution

The recommended solution describes the most desirable way to solve the problem or leverage the opportunity. The solution is described in sufficient detail for decision makers to understand the solution and determine if the recommendation will be implemented. The recommended solution may also include some estimates of cost and duration to implement the solution. Measurable benefits/outcomes will be identified to allow stakeholders to assess the performance and success of the solution after implementation and during operation.

10.7.4 Usage Considerations

.1 Strengths

- Provides an amalgamation of the complex facts, issues, and analysis required to make decisions regarding change.
- Provides a detailed financial analysis of cost and benefits.
- Provides guidance for ongoing decision making throughout the initiative.

.2 Limitations

- May be subject to the biases of authors.
- Frequently not updated once funding for the initiative is secured.
- Contains assumptions regarding costs and benefits that may prove invalid upon further investigation.

10.8 Business Model Canvas

10.8.1 Purpose

A business model canvas describes how an enterprise creates, delivers, and captures value for and from its customers.

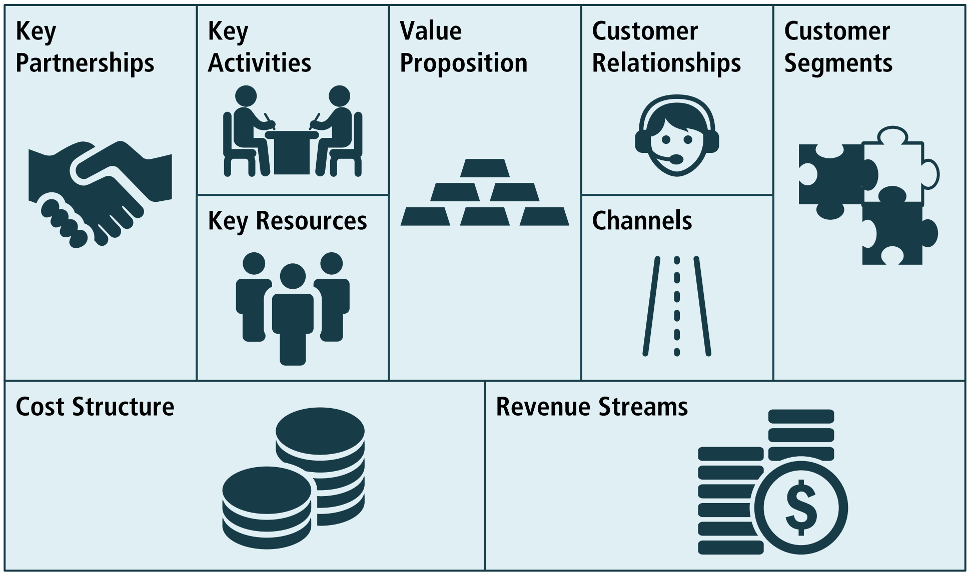
10.8.2 Description

A business model canvas is comprised of nine building blocks that describe how an organization intends to deliver value:

- Key Partnerships,
- Key Activities,
- Key Resources,
- Value Proposition,
- Customer Relationships,
- Channels,
- Customer Segments,
- Cost Structure, and
- Revenue Streams.

These building blocks are arranged on a business canvas that shows the relationship between the organization's operations, finance, customers, and offerings. The business model canvas also serves as a blueprint for implementing a strategy.

Figure 10.8.1: Business Model Canvas



A business model canvas can be used as a diagnostic and planning tool regarding strategy and initiatives. As a diagnostic tool, the various elements of the canvas are used as a lens into the current state of the business, especially with regards to the relative amounts of energy, time, and resources the organization is currently investing in various areas. As a planning and monitoring tool, the canvas can be used as a guideline and framework for understanding inter-dependencies and priorities among groups and initiatives.

A business model canvas allows for the mapping of programs, projects, and other initiatives (such as recruitment or talent retention) to the strategy of the enterprise. In this capacity, the canvas can be used to view where the enterprise is investing, where a particular initiative fits, and any related initiatives.

A business model canvas can also be used to demonstrate where the efforts of various departments and work groups fit and align to the overall strategy of the enterprise.

.1 Elements

Key Partnerships

Key partnerships frequently involve some degree of sharing of proprietary information, including technologies. An effective key partnership can, in some cases, lead to more formalized relationships such as mergers and acquisitions.

The benefits in engaging in key partnerships include:

- optimization and economy,
- reduction of risk and uncertainty,
- acquisition of particular resources and activities, and
- lack of internal capabilities.

Key Activities

Key activities are those that are critical to the creation, delivery, and maintenance of value, as well as other activities that support the operation of the enterprise.

Key activities can be classified as:

- **Value-add**: characteristics, features, and business activities for which the customer is willing to pay.
- **Non-value-add**: aspects and activities for which the customer is not willing to pay.
- **Business non-value-add**: characteristics that must be included in the offering, activities performed to meet regulatory and other needs, or costs associated with doing business, for which the customer is not willing to pay.

Key Resources

Resources are the assets needed to execute a business model. Resources may be different based on the business model.

Resources can be classified as:

- **Physical**: applications, locations, and machines.
- **Financial**: what is needed to fund a business model, such as cash and lines of credit.
- **Intellectual**: any proprietary aspects that enable a business model to thrive, such as knowledge, patents and copyrights, customer databases, and branding.
- **Human**: the people needed to execute a particular business model.

Value Proposition

A value proposition represents what a customer is willing to exchange for having their needs met. The proposition may consist of a single product or service, or may be comprised of a set of goods and services that are bundled together to address the needs of a customer or customer segment to help them solve their problem.

Customer Relationships

In general, customer relationships are classified as customer acquisition and customer retention. The methods used in establishing and maintaining customer relationships vary depending on the level of interaction desired and the method of communication. For example, some relationships can be highly personalized, while others are automated and promote a self-serve approach. The relationships can also be formal or informal.

Organizations interact with their customers in different ways depending on the relationship they want to establish and maintain.

Channels

Channels are the different ways an enterprise interacts with and delivers value to its customers. Some channels are very communication-oriented (for example, marketing channel), and some are delivery-oriented (for example, distribution channel). Other examples include sales channels and partnering channels.

Enterprises use channels to:

- raise awareness about their offerings,
- help customers evaluate the value proposition,
- allow customers to purchase a good or service,
- help the enterprise deliver on the value proposition, and
- provide support.

Understanding channels involves identifying the processes, procedures, technologies, inputs, and outputs (and their current impact), as well as understanding the relationship of the various channels to the strategies of the organization.

Customer Segments

Customer segments group customers with common needs and attributes so that the enterprise can more effectively and efficiently address the needs of each segment.

An organization within an enterprise may consider defining and targeting distinct customer segments based on:

- different needs for each segment,
- varying profitability between segments,
- different distribution channels, and
- formation and maintenance of customer relationships.

Cost Structure

Every entity, product, or activity within an enterprise has an associated cost. Enterprises seek to reduce, minimize, or eliminate costs wherever possible. Reducing costs may increase the profitability of an organization and allow those funds to be used in other ways to create value for the organization and for customers. Therefore, it is important to understand the type of business models, the differences in the types of costs and their impact, and where the enterprise is focusing its efforts to reduce costs.

Revenue Streams

A revenue stream is a way or method by which revenue comes into an enterprise from each customer segment in exchange for the realization of a value proposition. There are two basic ways revenue is generated for an enterprise:

revenue resulting from a one-time purchase of a good or service and recurring revenue from periodic payments for a good, service, or ongoing support.

Some types of revenue streams include:

- **Licensing or Subscription fees**: the customer pays for the right to access a particular asset, either as a one-time fee or as a recurring cost.
- **Transaction or Usage fees**: the customer pays each time they use a good or service.
- **Sales**: the customer is granted ownership rights to a specific product.
- **Lending, Renting, or Leasing**: the customer has temporary rights to use an asset.

.2 Usage Considerations

Strengths

- It is a widely used and effective framework that can be used to understand and optimize business models.
- It is simple to use and easy to understand.

Limitations

- Does not account for alternative measures of value such as social and environmental impacts.
- The primary focus on value propositions does not provide a holistic insight for business strategy.
- Does not include the strategic purpose of the enterprise within the canvas.

10.9 Business Rules Analysis

10.9.1 Purpose

Business rules analysis is used to identify, express, validate, refine, and organize the rules that shape day-to-day business behaviour and guide operational business decision making.

10.9.2 Description

Business policies and rules guide the day-to-day operation of the business and its processes, and shape operational business decisions. A business policy is a directive concerned with broadly controlling, influencing, or regulating the actions of an enterprise and the people in it. A business rule is a specific, testable directive that serves as a criterion for guiding behaviour, shaping judgments, or making decisions. A business rule must be practicable (needing no further interpretation for use by people in the business) and is always under the control of the business.

Analysis of business rules involves capturing business rules from sources, expressing them clearly, validating them with stakeholders, refining them to best align with business goals, and organizing them so they can be effectively managed and reused. Sources of business rules may be explicit (for example, documented business policies, regulations, or contracts) or tacit (for example, undocumented stakeholder know-how, generally accepted business practices, or norms of the corporate culture). Business rules should be explicit, specific, clear, accessible, and single sourced. Basic principles for business rules include:

- basing them on standard business vocabulary to enable domain subject matter experts to validate them,
- expressing them separately from how they will be enforced,
- defining them at the atomic level and in declarative format,
- separating them from processes they support or constrain,
- mapping them to decisions the rule supports or constrains, and
- maintaining them in a manner such that they can be monitored and adapted as business circumstances evolve over time.

A set of rules for making an operational business decision may be expressed as a decision table or decision tree, as described in Decision Analysis (p. 261). The number of rules in such a set can be quite large, with a high level of complexity.

10.9.3 Elements

Business rules require consistent use of business terms, a glossary of definitions for the underlying business concepts, and an understanding of the structural connections among the concepts. Reuse of existing terminology from external industry associations or internal business glossaries is often advised. Sometimes definitions and structures from data dictionaries or data models can be helpful (see Data Dictionary (p. 247) and Data Modelling (p. 256)). Business rules should be expressed and managed independently of any implementation technology since they need to be available for reference by business people. In addition, they sometimes will be implemented in multiple platforms or software components. There are frequently exceptions to business rules; these should be treated simply as additional business rules. Existing business rules should be challenged to ensure they align with business goals and remain relevant, especially when new solutions emerge.

.1 Definitional Rules

Definitional rules shape concepts, or produce knowledge or information. They indicate something that is necessarily true (or untrue) about some concept, thereby supplementing its definition. In contrast to behavioural rules, which are about the behaviour of people, definitional rules represent operational knowledge of the organization. Definitional rules cannot be violated but they can be misapplied. An example of a definitional rule is:

> *A customer must be considered a Preferred Customer if they place more than 10 orders per month.*

Definitional rules often prescribe how information may be derived, inferred or calculated based on information available to the business. An inference or calculation may be the result of multiple rules, each building on something inferred or calculated by some other(s). Sets of definitional rules are often used to make operational business decisions during some process or upon some event. An example of a calculation rule is:

> *An order's local jurisdiction tax amount must be calculated as (sum of the prices of all the order's taxable ordered items) × local jurisdiction tax rate amount.*

.2 Behavioural Rules

Behavioural rules are people rules–even if the behaviour is automated. Behavioural rules serve to shape (govern) day-to-day business activity. They do so by placing some obligation or prohibition on conduct, action, practice, or procedure.

Behavioural rules are rules the organization chooses to enforce as a matter of policy, often to reduce risk or enhance productivity. They frequently make use of the information or knowledge produced by definitional rules (which are about shaping knowledge or information). Behavioural rules are intended to guide the actions of people working within the organization, or people who interact with it. They may oblige individuals to perform actions in a certain way, prevent them from carrying out actions, or prescribe the conditions under which something can be correctly done. An example of a behavioural rule is:

> *An order must not be placed when the billing address provided by the customer does not match the address on file with the credit card provider.*

In contrast to definitional rules, behavioural rules are rules that can be violated directly. By definition, it is always possible to violate a behavioural rule—even if there are no circumstances under which the organization would approve that, and despite the fact that the organization takes extraordinary precautions in its solution to prevent it. Because of this, further analysis should be conducted to determine how strictly the rule needs to be enforced, what kinds of sanctions should be imposed when it is violated, and what additional responses to a violation might be appropriate. Such analysis often leads to specification of additional rules.

Various levels of enforcement may be specified for a behavioural rule. For example:

- Allow no violations (strictly enforced).
- Override by authorized actor.

- Override with explanation.
- No active enforcement.

A behavioural rule for which there is no active enforcement is simply a guideline that suggests preferred or optimal business behaviour.

10.9.4 Usage Considerations

.1 Strengths

- When enforced and managed by a single enterprise-wide engine, changes to business rules can be implemented quickly.
- A centralized repository creates the ability to reuse business rules across an organization.
- Business rules provide structure to govern business behaviours.
- Clearly defining and managing business rules allows organizations to make changes to policy without altering processes or systems.

.2 Limitations

- Organizations may produce lengthy lists of ambiguous business rules.
- Business rules can contradict one another or produce unanticipated results when combined unless validated against one another.
- If available vocabulary is insufficiently rich, not business-friendly, or poorly defined and organized, resulting business rules will be inaccurate or contradictory.

10.10 Collaborative Games

10.10.1 Purpose

Collaborative games encourage participants in an elicitation activity to collaborate in building a joint understanding of a problem or a solution.

10.10.2 Description

Collaborative games refer to several structured techniques inspired by game play and are designed to facilitate collaboration. Each game includes rules to keep participants focused on a specific objective. The games are used to help the participants share their knowledge and experience on a given topic, identify hidden assumptions, and explore that knowledge in ways that may not occur during the course of normal interactions. The shared experience of the collaborative game encourages people with different perspectives on a topic to work together in order to better understand an issue and develop a shared model of the problem or of potential solutions. Many collaborative games can be used to understand the perspectives of various stakeholder groups.

Collaborative games often benefit from the involvement of a neutral facilitator who helps the participants understand the rules of the game and enforces those rules. The facilitator's job is to keep the game moving forward and to help ensure that all participants play a role. Collaborative games usually involve a strong visual or tactile element. Activities such as moving sticky notes, scribbling on whiteboards, or drawing pictures help people to overcome inhibitions, foster creative thinking, and think laterally.

10.10.3 Elements

.1 Game Purpose

Each different collaborative game has a defined purpose—usually to develop a better understanding of a problem or to stimulate creative solutions—that is specific to that type of game. The facilitator helps the participants in the game understand the purpose and work toward the successful realization of that purpose.

.2 Process

Each type of collaborative game has a process or set of rules that, when followed, keeps the game moving toward its goal. Each step in the game is often limited by time.

Games typically have at least three steps:

Step 1. an opening step, in which the participants get involved, learn the rules of the game, and start generating ideas,

Step 2. the exploration step, in which participants engage with one another and look for connections between their ideas, test those ideas, and experiment with new ideas, and

Step 3. a closing step, in which the ideas are assessed and participants work out which ideas are likely to be the most useful and productive.

.3 Outcome

At the end of a collaborative game, the facilitator and participants work through the results and determine any decisions or actions that need to be taken as a result of what the participants have learned.

.4 Examples of Collaborative Games

There are many types of collaborative games available, including (but not limited to) the following:

Table 10.10.1: Examples of Collaborative Games

Game	Description	Objective
Product Box	Participants construct a box for the product as if it was being sold in a retail store.	Used to help identify features of a product that help drive interest in the marketplace.
Affinity Map	Participants write down features on sticky notes, put them on a wall, and then move them closer to other features that appear similar in some way.	Used to help identify related or similar features or themes.
Fishbowl	Participants are divided into two groups. One group of participants speaks about a topic, while the other group listens intently and documents their observations.	Used to identify hidden assumptions or perspectives.

10.10.4 Usage Considerations

.1 Strengths

- May reveal hidden assumptions or differences of opinion.
- Encourages creative thinking by stimulating alternative mental processes.
- Challenges participants who are normally quiet or reserved to take a more active role in team activities.
- Some collaborative games can be useful in exposing business needs that aren't being met.

.2 Limitations

- The playful nature of the games may be perceived as silly and make participants with reserved personalities or cultural norms uncomfortable.
- Games can be time-consuming and may be perceived as unproductive, especially if the objectives or outcomes are unclear.
- Group participation can lead to a false sense of confidence in the conclusions reached.

10.11 Concept Modelling

10.11.1 Purpose

A concept model is used to organize the business vocabulary needed to consistently and thoroughly communicate the knowledge of a domain.

10.11.2 Description

A concept model starts with a glossary, which typically focuses on the core noun concepts of a domain. Concept models put a premium on high-quality, design-independent definitions that are free of data or implementation biases. Concept models also emphasize rich vocabulary.

A concept model identifies the correct choice of terms to use in communications, including all business analysis information. It is especially important where high precision and subtle distinctions need to be made.

Concept models can be effective where:

- the enterprise seeks to organize, retain, build-on, manage, and communicate core knowledge,
- the initiative needs to capture large numbers of business rules,
- there is resistance from stakeholders about the perceived technical nature of data models, class diagrams, or data element nomenclature and definition,
- innovative solutions are sought when re-engineering business processes or other aspects of business capability, and
- the enterprise faces regulatory or compliance challenges.

A concept model differs from a data model. The goal of a concept model is to support the expression of natural language statements, and supply their semantics. Concept models are not intended to unify, codify, and simplify data. Therefore the vocabulary included in a concept model is far richer, as suits knowledge-intensive domains. Concept models are often rendered graphically.

10.11.3 Elements

.1 Noun Concepts

The most basic concepts in a concept model are the noun concepts of the domain, which are simply ‘givens’ for the space.

.2 Verb Concepts

Verb concepts provide basic structural connections between noun concepts. These verb concepts are given standard wordings, so they can be referenced unambiguously. These wordings by themselves are not necessarily sentences; rather, they are the building blocks of sentences (such as business rule statements). Sometimes verb concepts are derived, inferred, or computed by definitional rules. This is how new knowledge or information is built up from more basic facts.

.3 Other Connections

Since concept models must support rich meaning (semantics), other types of standard connections are used besides verb concepts.

These include but are not limited to:

- categorizations,
- classifications,
- partitive (whole-part) connections, and
- roles.

10.11.4 Usage Considerations

.1 Strengths

- Provide a business-friendly way to communicate with stakeholders about precise meanings and subtle distinctions.
- Is independent of data design biases and the often limited business vocabulary coverage of data models.
- Proves highly useful for white-collar, knowledge-rich, decision-laden business processes.
- Helps ensure that large numbers of business rules and complex decision tables are free of ambiguity and fit together cohesively.

.2 Limitations

- May set expectations too high about how much integration based on business semantics can be achieved on relatively short notice.
- Requires a specialized skill set based on the ability to think abstractly and non-procedurally about know-how and knowledge.
- The knowledge-and-rule focus may be foreign to stakeholders.
- Requires tooling to actively support real-time use of standard business terminology in writing business rules, requirements, and other forms of business communication.

10.12 Data Dictionary

10.12.1 Purpose

A data dictionary is used to standardize a definition of a data element and enable a common interpretation of data elements.

10.12.2 Description

A data dictionary is used to document standard definitions of data elements, their meanings, and allowable values. A data dictionary contains definitions of each data element and indicates how those elements combine into composite data elements. Data dictionaries are used to standardize usage and meanings of data elements between solutions and between stakeholders.

Data dictionaries are sometimes referred to as metadata repositories and are used to manage the data within the context of a solution. As organizations adopt data mining and more advanced analytics, a data dictionary may provide the metadata required by these more complex scenarios. A data dictionary is often used in conjunction with an entity relationship diagram (see Data Modelling (p. 256)) and may be extracted from a data model.

Data dictionaries can be maintained manually (as a spreadsheet) or via automated tools.

Figure 10.12.1: Example of a Data Dictionary

Primitive Data Elements	Data Element 1	Data Element 2	Data Element 3
Name Name referenced by data elements	First Name	Middle Name	Last Name
Alias Alternate name referenced by stakeholders	Given Name	Middle Name	Surname
Values/Meanings Enumerated list or description of data element	Minimum 2 characters	Can be omitted	Minimum 2 characters
Description Definition	First Name	Middle Name	Family Name
Composite	**Customer Name =** First Name + Middle Name + Family Name		

10.12.3 Elements

.1 Data Elements

Data dictionaries describe data element characteristics including the description of the data element in the form of a definition that will be used by stakeholders. Data dictionaries include standard definitions of data elements, their meanings, and allowable values. A data dictionary contains definitions of each primitive data element and indicates how those elements combine into composite data elements.

.2 Primitive Data Elements

The following information must be recorded about each data element in the data dictionary:

- **Name**: a unique name for the data element, which will be referenced by the composite data elements.
- **Aliases**: alternate names for the data element used by various stakeholders.
- **Values/Meanings**: a list of acceptable values for the data element. This may be expressed as an enumerated list or as a description of allowed formats for the data (including information such as the number of characters). If the values are abbreviated this will include an explanation of the meaning.
- **Description**: the definition of the data element in the context of the solution.

.3 Composite Elements

Composite data elements are built using data elements to build composite structures, which may include:

- **Sequences**: required ordering of primitive data elements within the composite structure. For example, a plus sign indicates that one element is followed by or concatenated with another element: Customer Name = First Name+Middle Name+Family Name.
- **Repetitions**: whether one or more data elements may be repeated multiple times.
- **Optional Elements**: may or may not occur in a particular instance of the composite element.

10.12.4 Usage Considerations

.1 Strengths

- Provides all stakeholders with a shared understanding of the format and content of relevant information.
- A single repository of corporate metadata promotes the use of data throughout the organization in a consistent manner.

.2 Limitations

- Requires regular maintenance, otherwise the metadata could become obsolete or incorrect.
- All maintenance is required to be completed in a consistent manner in order to ensure that stakeholders can quickly and easily retrieve the information they need. This requires time and effort on the part of the stewards responsible for the accuracy and completeness of the data dictionary.
- Unless care is taken to consider the metadata required by multiple scenarios, it may have limited value across the enterprise.